

Career Operations Intelligence Platform

Overview

The Career Operations Intelligence Platform was developed in response to a practical problem: the volume of incoming job opportunities had grown beyond what could be consistently evaluated by hand.

Job alerts, recruiter outreach, company notifications, networking referrals, and job-board subscriptions were generating dozens of opportunities each week. Opportunity availability was not the problem. The challenge was the time required to review, compare, prioritize, and act on those opportunities in a consistent and rational manner.

The project began as an attempt to create a more structured process for evaluating opportunities. Rather than relying on ad hoc review of individual postings, the goal was to build a system capable of collecting opportunities from multiple sources, converting them into a common format, evaluating them against predefined criteria, and presenting a prioritized set of opportunities for review.

As development progressed, however, the project became less about job-search automation and more about building reliable AI-assisted workflows.

Early prototypes revealed limitations that are common in large language model systems. Models frequently introduced unsupported assumptions, generated inconsistent evaluations, failed to preserve source information, and occasionally produced outputs that appeared plausible but could not be traced back to underlying data.

These observations shifted the focus of the project. The primary challenge was no longer generating recommendations. The challenge was producing recommendations that remained explainable, traceable, and predictable.

Much of the platform architecture emerged directly from this realization. Deterministic processes were introduced for extraction, normalization, deduplication, provenance tracking, schema validation, reporting, and storage. AI systems were reserved for tasks involving judgment, prioritization, and comparative evaluation.

The resulting platform functions as a supervised decision-support system rather than an autonomous AI application. Human judgment remains responsible for all consequential decisions, while AI is used selectively to assist with evaluation, prioritization, and analysis.

Design Principles

Several principles guided the design of the platform.

The first is that opportunity volume is not inherently valuable. Many job-search tools focus on finding more opportunities and increasing application volume. This platform focuses on identifying which opportunities deserve attention.

The second is that interview probability matters more than aspirational matching. A realistic bridge opportunity is often more valuable than a prestigious role that is unlikely to generate a first-round interview.

The third is that deterministic tasks and judgment tasks should be separated whenever possible. Repeatable operations such as extraction, normalization, storage, reporting, and validation are handled through deterministic workflows. AI systems are used where ambiguity and tradeoffs require evaluation.

Finally, recommendations must remain explainable. Outputs should be traceable to source information and accompanied by sufficient context to support human review.

Technical Architecture

The platform was designed to run on consumer hardware using tools that were already available within the existing workflow environment.

Current components include:

- ❖ Docker Desktop
- ❖ n8n
- ❖ Gmail API
- ❖ Google Gemini
- ❖ Obsidian
- ❖ Ollama
- ❖ Qwen 2.5 14B

The architecture intentionally favors maintainability and transparency over scale.

Rather than building a monolithic AI application, the platform combines workflow automation, deterministic processing, and AI-assisted reasoning into separate layers.

System Architecture

The platform consists of two independent workflows.

Workflow A: Career Operations Intelligence

Workflow A serves as the strategic intelligence layer.

Its purpose is to answer a single question:

"Which opportunities deserve attention?"

The workflow collects opportunity signals from multiple sources, converts them into structured records, removes duplicates, evaluates opportunities against a defined career strategy, and generates ranked recommendations.

The workflow performs:

- ❖ opportunity ingestion
- ❖ extraction
- ❖ normalization
- ❖ deduplication
- ❖ opportunity evaluation
- ❖ ranking
- ❖ reporting

The output is a daily intelligence briefing that summarizes the most promising opportunities identified during the evaluation cycle.

Workflow B: Resume Routing & Terminology Intelligence

Workflow B serves as the tactical application-support layer.

Its purpose is to answer a different question:

"How should this opportunity be approached?"

Once an opportunity has been selected for review, Workflow B evaluates the job description against a portfolio of resume profiles and identifies terminology that may warrant manual consideration before submission.

The workflow performs:

- ❖ resume routing

- ❖ terminology analysis
- ❖ confidence assessment
- ❖ report generation

Workflow B does not generate resumes, rewrite experience, or submit applications.

Together they create a structured decision process:

Opportunity Discovery → Opportunity Evaluation → Opportunity Prioritization → Resume Routing → Human Review → Application Decision

This separation reduces complexity, improves maintainability, and allows strategic evaluation and application preparation to evolve independently.

Decision Framework

The platform evaluates opportunities using multiple dimensions rather than relying solely on title matching.

Evaluation considers:

- ❖ interview probability
- ❖ career capital
- ❖ long-term optionality
- ❖ organizational development value
- ❖ geographic practicality
- ❖ compensation

The framework assumes that opportunities cannot generate value if they are unlikely to produce an interview.

As a result, interview probability functions as an important constraint throughout the evaluation process.

The system generally favors opportunities that:

- ❖ build transferable experience
- ❖ increase future career options
- ❖ provide meaningful professional development
- ❖ align with long-term career objectives
- ❖ remain realistically attainable

This often results in bridge opportunities receiving higher rankings than destination roles.

Lessons Learned

One of the most valuable outcomes of the project was a deeper understanding of the strengths and limitations of modern language models.

The models consistently performed well when asked to:

- ❖ compare alternatives
- ❖ identify patterns
- ❖ classify opportunities
- ❖ summarize information
- ❖ provide qualitative assessments

They performed less reliably when asked to:

- ❖ preserve strict structure
- ❖ maintain complete traceability
- ❖ execute deterministic logic
- ❖ enforce schema consistency
- ❖ process large datasets without constraints

Many architectural decisions emerged directly from these observations.

Confidence scoring, structured JSON outputs, schema validation, canonicalization layers, workflow separation, and human review checkpoints were all introduced to reduce the impact of hallucinations, inconsistency, and information loss.

In practice, the project demonstrated that effective AI systems often depend less on model sophistication and more on careful workflow design.

Scope

The platform currently supports:

- ❖ opportunity ingestion
- ❖ extraction
- ❖ normalization
- ❖ deduplication
- ❖ evaluation

- ❖ prioritization
- ❖ resume routing
- ❖ terminology analysis
- ❖ reporting

The platform does not support:

- ❖ automated application submission
- ❖ resume generation
- ❖ cover-letter generation
- ❖ interview scheduling
- ❖ candidate representation
- ❖ autonomous decision making

These exclusions are deliberate. The objective is to improve decision quality rather than maximize automation.

Human Oversight

Human review remains embedded throughout the platform.

The system may:

- ❖ recommend opportunities
- ❖ recommend resumes
- ❖ identify terminology
- ❖ assign confidence scores
- ❖ generate prioritization guidance

The system may not:

- ❖ submit applications
- ❖ alter candidate history
- ❖ fabricate qualifications
- ❖ override human judgment

Responsibility for resume edits, application submission, networking activities, and career decisions remains entirely with the applicant.

Conclusion

The Career Operations Intelligence Platform ultimately became less about automating a job search and more about building a practical framework for human-AI collaboration. What began as an effort to manage a growing volume of job opportunities evolved into an exploration of reliability, explainability, and workflow design. The resulting system demonstrates how AI can be incorporated into decision-support processes while preserving transparency, traceability, and human control. The platform is therefore best understood not as an autonomous career tool, but as a structured intelligence system designed to support better decisions in an increasingly noisy information environment.